
MUHAMMAD RESHAEEL



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Address: Abu Dhabi, United Arab Emirates (UAE)

Gender: Male

Date of birth: Oct 18th, 1996

Nationality: Pakistani

Professional Summary

I hold a PhD in Mechanical Engineering with expertise in modelling, simulation, and data-driven analysis of energy systems, particularly HVAC and advanced thermal systems. My research is focused on physics-informed machine learning (grey-box modelling) for energy prediction, system optimization, and fault detection and diagnostics. I have experience in analyzing dynamic behavior of energy systems to improve performance and reliability, supported by a strong foundation in thermodynamics and system-level modelling. My work combines first-principles physics with data-driven methods, supported by proficiency in Python, MATLAB-based modeling, and machine/deep learning techniques for energy applications.

Research Interests

Physics-informed machine and deep learning for energy system modeling | Energy prediction | Fault detection and diagnostics | Modelling and optimization of cooling systems, buildings, and advanced power systems | Physics-guided and data-driven analysis of thermodynamic behavior of energy systems | Waste heat recovery and repowering of power systems | Building energy performance and system-level energy analysis

Education

1.) Degree: PhD in Engineering-Mechanical Engineering

Tenure: September 2022 – April 2026

Institute: Khalifa University, Abu Dhabi, UAE

CGPA: 3.79/4.00

Thesis: Advanced Cooling Technology for Building Energy Efficiency

2.) Degree: Master of Science in Thermal Energy Engineering

Tenure: September 2019 – November 2021

Institute: National University of Sciences and Technology (NUST), Islamabad, Pakistan

CGPA: 4.00/4.00 (Gold medalist)

Thesis: Parametric Optimization of a Double Stage Reheated organic Rankine cycle (ORC) for Recovering Waste Heat from a Combined Cycle Gas Turbine Power Plant in the Perspective of Repowering

3.) Degree: Bachelors of Science in Mechanical Engineering

Tenure: September 2015 – September 2019

Institute: University of Engineering and Technology, Peshawar, Pakistan

CGPA: 3.65/4.00 (With distinction)

Thesis: Comparative study of Carbon Capturing Techniques with Design and Fabrication of a Prototype Model

Work Experience

1.) Position: Graduate Teaching Assistant

Tenure: Sep 2022 - Current

Institute: Khalifa University, Abu Dhabi, UAE

Key responsibilities:

- Assistance in teaching graduate courses.
- Conduction of lab activities.

2.) Position: Lecturer

Tenure: Feb 2022 – Aug 2022

Institute: University of Wah, Wah Cantt, Pakistan

Key responsibilities:

- Taught the following subjects: Thermodynamics and heat transfer.
- Academic planning in the perspective of Objective-Based Education (OBE).

3.) Position: Lab Engineer

Tenure: Feb 2021 – Feb 2022

Institute: CECOS University of IT and Emerging Sciences, Peshawar, Pakistan

Key responsibilities:

- Taught lab of power plants and subject of Engineering thermodynamics.
- Maintenance of the mechanical equipment.
- Academic planning in the perspective of Objective-Based Education (OBE).

4.) Position: Management Associate Engineer

Tenure: Sep 2020 – Dec 2021

Company: Fauji Fertilizers Bin Qasim Limited, Pakistan

Key responsibilities:

- Maintenance of heat exchangers.
- Cooling tower load calculation

List of Publications

1.) Title: A Hybrid Autoencoder-Random Forest Framework for Open-Set Recognition of Sensor and Communication Faults in a Variable Refrigerant Flow System with Real-Time Deployment Analysis

Status: Published (DOI to be issued)

Authors: Muhammad Reshaeel¹, Mohamed I. Hassan Ali^{1*}

Journal: Energy

Ranking: Top 1% (IF 9.4)

2.) Title: Physics-guided, noise-resilient and interpretable machine learning framework for fault detection and diagnosis in variable refrigerant flow systems

DOI: <https://doi.org/10.1016/j.egyai.2026.100766>

Authors: Muhammad Reshaeel¹, Vinod Khadkikar², Mohamed I. Hassan Ali^{1*}

Journal: Energy and AI

Ranking: Top 4% (IF 9.6)

3.) Title: Enhancing energy prediction in variable refrigerant flow cooling systems: a hybrid grey-box model integrating thermodynamic knowledge, experimental data and machine learning
DOI: <https://doi.org/10.1016/j.energy.2025.139895>
Authors: Muhammad Reshaeel¹, Mohamed I. Hassan Ali^{1*}
Journal: Energy
Ranking: Top 1% (IF 9.4)

4.) Title: Concentrated solar power (CSP) driven desalination systems: A techno-economic review
DOI: <https://doi.org/10.1016/j.rser.2025.116311>
Authors: M. Imran Khan^a, Muhammad Reshaeel^b, Faisal Asfand^c, Sami G. Al-Ghamdi^d, Muhammad Farooq^a, Mushtaq Khan^a, Furqan Tahir^d, Yusuf Bicer^e, Muhammad Asif^{fi}, Mohammad Rehan^g, Tonni Agustiono Kurniawan^h
Journal: Renewable and Sustainable Energy Reviews
Ranking: Top 2% (IF 16.3)

5.) Title: Inherent thermodynamic performance assessment of a variable refrigerant flow system under transient cooling load: A case study of an eco-villa
DOI: <https://doi.org/10.1016/j.csite.2025.106171>
Authors: Muhammad Reshaeel¹, Mohamed I. Hassan Ali^{1*}
Journal: Case Studies in Thermal Engineering
Ranking: Top 8% (IF 6.4)

6.) Title: A critical review of the thermal-hydraulic performance of fin and tube heat exchangers using statistical analysis
DOI: <https://doi.org/10.1016/j.ijft.2024.100858>
Authors: Muhammad Reshaeel^a, Mostafa M. Abdelsamie^{a b 1}, Mohamed I. Hassan Ali^{a*}
Journal: International Journal of Thermofluids
Ranking: Top 10%

7.) Title: Multiparametric optimization of a reheated organic Rankine cycle for waste heat recovery based repowering of a degraded combined cycle gas turbine power plant
DOI: <https://doi.org/10.1016/j.enconman.2022.115237>
Authors: Muhammad Reshaeel, Adeel Javed^{*}, Ahmad Jamil, Majid Ali, Mariam Mahmood, Adeel Waqas
Journal: Energy Conversion and Management
Ranking: Top 2% (IF 10.9)

Skills

Softwares

Python | MATLAB | Hourly Analysis Program (HAP) | Cycle Tempo | System Advisory Model (SAM) | ANSYS | Design Expert

Experimental and research skills

Physics-informed machine and deep learning for energy prediction | Energy modeling and optimization (HVAC, building systems and advanced thermodynamic cycles) | Physics-guided machine learning framework development for fault detection and diagnostics | HVAC test bench development | Instrument calibration | Uncertainty quantification | Exergy and CO₂ emission analysis | Statistical analysis | Technoeconomic analysis

Honors, Awards and Certifications

- **Graduate Research Award**

Awarding institute: Khalifa University

- **Doctoral Research Scholarship Award**

Awarding institute: Khalifa University

- **President Gold Medal Award in Master of Science in Thermal Energy Engineering**

Awarding institute: National University of Sciences and Technology (NUST)

- **Distinction in Bachelor of Science in Mechanical Engineering**

Awarding institute: University of Engineering and Technology (UET)

- **Best Final Year Project in Environment Category**

Awarding institute: University of Engineering and Technology (UET)

- **Wind Energy Course**

Awarding institute: Technical University of Denmark (DTU)

- **Introduction to Programming with MATLAB**

Awarding institute: Vanderbilt University

Languages

- **Mother tongue(s):** Urdu

- **Other languages:** English speaking (advanced), reading (advanced), listening (advanced), writing (advanced) **TOEFL iBT:** 108/120

Recommendations

1.) Name: Dr. Mohamed I. Hassan Ali

Email: mohamed.ali@ku.ac.ae

Designation: Associate Professor

Department: Mechanical and Nuclear Engineering

Institute: Khalifa University

Relation: Advisor

2.) Name: Dr. Dimitris Goussis

Email: dimitris.goussis@ku.ac.ae

Designation: Professor

Department: Mechanical and Nuclear Engineering

Institute: Khalifa University

Relation: RSC member

3.) Name: Dr. Majid Ali

Email: majid@uspcase.nust.edu.pk

Designation: Head of Department | Associate Professor

Department: Thermal Energy Engineering

Institute: US Pakistan Center for Advanced Studies in Energy (USPCAS-E), National University of Sciences and Technology (NUST)

Relation: Co-Advisor